WORKED SOLUTION

STEP III 1990, Question 14

Mechanics • Systems of Forces

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Read the question before you go on. It is in the STEP Support Programme's question database:

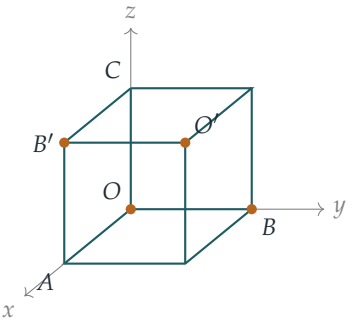
step.maths.org/questions/90-s3-q14

The key idea

A system of forces on a rigid body comes down to two vectors: the resultant $\mathbf{R} = \sum \mathbf{F}_i$ and the moment about a chosen point, $\mathbf{G}_P = \sum(\mathbf{r}_i - \mathbf{p}) \times \mathbf{F}_i$. Moving the base point from O to P changes the moment by a simple rule, $\mathbf{G}_P = \mathbf{G}_O - \mathbf{p} \times \mathbf{R}$. Every part of the question is this rule used with a different P.

Solution

Model the cube as a rigid body with edge a, so $A = (a, 0, 0)$, $B = (0, a, 0)$, $C = (0, 0, a)$, and the vertex opposite B is $B' = (a, 0, a)$.



Finding λ, μ, ν

The force at O has no moment about O, so only the other three count. Using $(x,y,z)\times(p,q,r)=(yr-zq,\ zp-xr,\ xq-yp)$:

$$\overrightarrow{OO'}\times\begin{pmatrix}\lambda\\0\\1\end{pmatrix}=a\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}\lambda\\0\\1\end{pmatrix}=a\begin{pmatrix}1\\\lambda-1\\-\lambda\end{pmatrix},$$

$$\overrightarrow{OB}\times\begin{pmatrix}-1\\0\\2\end{pmatrix}=a\begin{pmatrix}0\\1\\0\end{pmatrix}\times\begin{pmatrix}-1\\0\\2\end{pmatrix}=a\begin{pmatrix}2\\0\\1\end{pmatrix},$$

$$\overrightarrow{OB'}\times\begin{pmatrix}1\\\mu\\\nu\end{pmatrix}=a\begin{pmatrix}1\\0\\1\end{pmatrix}\times\begin{pmatrix}1\\\mu\\\nu\end{pmatrix}=a\begin{pmatrix}-\mu\\1-\nu\\\mu\end{pmatrix}.$$

Adding, the total moment about O is $a\,(3-\mu,\ \lambda-\nu,\ 1-\lambda+\mu)$. Setting each component to zero gives $\mu=3$, then $\lambda=1+\mu=4$, then $\nu=\lambda=4$:

$$\lambda=4,\qquad \mu=3,\qquad \nu=4.$$

None of this depends on α,β,γ, and the resultant is

$$\mathbf{R}=\begin{pmatrix}\alpha+\lambda-1+1\\\beta+\mu\\\gamma+1+2+\nu\end{pmatrix}=\begin{pmatrix}\alpha+4\\\beta+3\\\gamma+7\end{pmatrix},\qquad \mathbf{G}_O=\mathbf{0}.$$

Part (i)

With $\alpha=\beta=\gamma=0$ we have $\mathbf{R}=(4,3,7)$. Two systems are equivalent when they have the same resultant and the same moment about one point (and hence about every point). So the single force at B must be $\mathbf{R}$ itself, and the couple must equal the moment of the original system about B:

$$\mathbf{G}_B=\mathbf{G}_O-\overrightarrow{OB}\times\mathbf{R}=\mathbf{0}-a\begin{pmatrix}0\\1\\0\end{pmatrix}\times\begin{pmatrix}4\\3\\7\end{pmatrix}=-a\begin{pmatrix}7\\0\\-4\end{pmatrix}=a\begin{pmatrix}-7\\0\\4\end{pmatrix}.$$

? **Ask yourself** Why is the moment about B equal to $\mathbf{G}_O-\overrightarrow{OB}\times\mathbf{R}$?

For each force, $(\mathbf{r}_i-\mathbf{b})\times\mathbf{F}_i=\mathbf{r}_i\times\mathbf{F}_i-\mathbf{b}\times\mathbf{F}_i$; summing over i gives $\mathbf{G}_O-\mathbf{b}\times\mathbf{R}$.

So the equivalent system is a force $(4,3,7)$ acting at B together with a couple of moment $a(-7,0,4)$. As a check, the force at B plus this couple has moment about O equal to $a(7,0,-4)+a(-7,0,4)=\mathbf{0}$, as it should.

Part (ii)

Now $\mathbf{R} = (2+4,\ 3+3,\ 2+7) = (6,6,9) = 3(2,2,3)$, and still $\mathbf{G}_O = \mathbf{0}$. At a point P with position vector $\mathbf{r} = (x,y,z)$,

$$\mathbf{G}_P = \mathbf{G}_O - \mathbf{r} \times \mathbf{R} = -\mathbf{r} \times \mathbf{R}.$$

This is zero exactly when $\mathbf{r}$ is parallel to $\mathbf{R}$ (or $\mathbf{r} = \mathbf{0}$), because $\mathbf{R} \neq \mathbf{0}$. So the locus is the line through O in the direction $(2,2,3)$:

$$\frac{x}{2} = \frac{y}{2} = \frac{z}{3}.$$

Physically, the whole system is equivalent to the single force $(6,6,9)$ acting along this line.

When the cube translates (no rotation), every point moves by the same displacement, so the work done by the forces is $\mathbf{R} \cdot \mathbf{s}$, where $\mathbf{s}$ is the displacement. For a distance d along the line, in the direction of $(2,2,3)$, $\mathbf{s} = \dfrac{d}{\sqrt{17}}(2,2,3)$ and

$$W = (6,6,9) \cdot \frac{d}{\sqrt{17}}(2,2,3) = \frac{51d}{\sqrt{17}} = 3\sqrt{17}\,d.$$

The question leaves the distance unnamed; per unit distance the work is $3\sqrt{17} \approx 12.37$ units. Since $\mathbf{R}$ lies along the line, this is just $|\mathbf{R}|\,d = \sqrt{36+36+81}\,d = \sqrt{153}\,d$, the same number. (Moving the other way along the line gives $-3\sqrt{17}\,d$.)

Checks

- With $\lambda = 4$, $\mu = 3$, $\nu = 4$, the three moments about O are $a(1,3,-4)$, $a(2,0,1)$ and $a(-3,-3,3)$, which sum to zero.
- In (ii), take $P = (2,2,3)$: $\mathbf{r} \times \mathbf{R} = (2,2,3) \times (6,6,9) = \mathbf{0}$. At $P = (1,0,0)$ the moment is $-(0,-9,6) \neq \mathbf{0}$.
- A computer with exact fractions confirmed every moment, checked that the equivalent system in (i) has the same moment as the original about 20 random points, and confirmed the work per unit distance is $3\sqrt{17} \approx 12.369$.

Common slips

- Getting the order wrong in $\mathbf{r} \times \mathbf{F}$: reversing it flips every sign.
- Taking the couple in (i) to be $+\overrightarrow{OB} \times \mathbf{R}$. Moving the force from O to B needs the compensating couple with the opposite sign.
- Finding the work by summing over the forces at different vertices. In a pure translation only the resultant matters.

Beyond STEP

Any system of forces on a rigid body reduces to a single force along a line together with a couple about that same line. This is a *wrench*, and its pitch is $\mathbf{R}\cdot\mathbf{G}/|\mathbf{R}|^2$. In (ii) the pitch is zero, so the wrench is just a single force, and the zero-moment locus is its line of action, the central axis. These ideas lead to rigid-body dynamics and screw theory in robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers torque and rigid bodies at the right level.

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